KSE Sentinel: A Rule-Based Trend-Continuation Strategy for the Pakistan Stock Exchange (PSX)

Design, Statistical Validation, and Live Forward Testing

Working Paper

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Abstract

I present KSE Sentinel, a fully rule-based, mechanically-defined trend-continuation strategy developed for the Pakistan Stock Exchange (PSX). The strategy formalizes a discretionary trading heuristic two consecutive same-direction candles closing on the trend-correct side of a short-period exponential moving average, with exit strictness conditioned on the prevailing trend regime into an auditable, mechanical rule set applied across a 30-symbol universe. We evaluate the strategy on 1,009 trades over an approximately five-year window (2021–2026), net of realistic transaction costs (commission, slippage, and spread totalling 22.5 basis points per leg), finding a 39.15% win rate, profit factor of 2.51, annualized Sharpe ratio of 2.08, and maximum drawdown of 2.91% in local currency terms. Consistent with the broader trend-following literature, returns are positively skewed: a minority of large winning trades drives the majority of profit, with average win size approximately 3.9 times average loss size . We address multiple-testing and selection bias directly using the Deflated Sharpe Ratio methodology of Bailey and López de Prado (2014), documenting an explicit floor on the number of prior trials considered . We further validate the strategy out-of-sample against an independent ~2000–2020 PSX historical archive, and distinct from purely backward-looking evidence report results from an ongoing 60-day live forward-paper validation program initiated July 2026, including real operational incidents caught and corrected during that window. We conclude with an honest discussion of currency risk (PKR-denominated returns are positive while USD-converted returns are negative over the sample period), capacity constraints, and the evolving Pakistani regulatory environment for algorithmic trading .

Keywords: trend following, momentum, frontier markets, Pakistan Stock Exchange, algorithmic trading, Deflated Sharpe Ratio, walk-forward validation, forward testing.

For Strategy and API link for testing click here https://shorturl.at/n9GTR

1 What This Paper Contributes

This paper does not propose a new theoretical mechanism for return predictability the underlying phenomenon (trend continuation / time-series momentum) is well documented across asset classes and decades . Instead, the contribution is applied and methodological:

2 A genuinely novel construction process. Rather than mining historical data for a profitable rule set the central concern of the backtest-overfitting literature KSE Sentinel's entry/exit logic was derived *first* from an experienced discretionary trader's codified judgment, and *only then* tested against historical data. The rules were not searched for; they were specified and then evaluated. This does not make the strategy immune to overfitting risk, but it is a materially different and arguably more defensible starting point than a data-mined rule.

3 Direct, transparent treatment of multiple-testing bias. Using the Deflated Sharpe Ratio , we report not just a headline Sharpe ratio but a statistically corrected assessment of how much of that Sharpe ratio survives once the number of prior strategy variants considered is accounted for and we disclose that trial count as an honest floor, not an exhaustive count (§6.3).

4 A genuine, ongoing live forward test, distinct from and reported separately from the backtest. Section 7 documents real operational incidents caught during live operation, presented transparently rather than smoothed over.

5 Honest treatment of currency risk for an international institutional audience: local-currency returns and USD-converted returns diverge meaningfully over the sample period, and we report both prominently.

2 Roadmap

Section 2 reviews the relevant literature. Section 3 describes the PSX market and the data used. Section 4 describes the strategy's methodology at a level appropriate for external review. Section 5 reports headline empirical results. Section 6 addresses statistical robustness and multiple-testing bias. Section 7 reports live forward-test results. Section 8 describes the operational and risk-management infrastructure. Section 9 states limitations plainly. Section 10 concludes.

2. Related Literature

2.1 Cross-Sectional and Time-Series Momentum

Jegadeesh and Titman (1993) established that strategies buying recent winners and selling recent losers generate significant positive returns over 3 - 12 month holding periods in U.S. equities. Moskowitz, Ooi, and Pedersen (2012) extended this to a timeseries ("trend") framework across 58 futures instruments spanning equities, currencies, commodities, and bonds, finding that past 12-month returns positively predict future returns . They document that this "time series momentum or 'trend' effect persists for about a year and then partially reverses over longer horizons," and that the findings are robust "across a number of sub-samples, look-back periods and holding periods" .

Hurst, Ooi, and Pedersen (2017) document that trend-following-style returns have persisted across approximately a century of data, across multiple asset classes and market regimes relevant context for why a mechanically-defined trend-continuation rule is not a speculative premise . Using 67 markets across four asset classes from 1880 to 2016, they found annualized gross returns of 18.0% with approximately half the volatility of equities, and net returns positive in every decade.

2.2 Positive Skew as a Signature of Trend-Following Returns

The broader literature on trend-following consistently documents a specific return signature: a high proportion of small losses offset by a smaller number of large gains, producing positive skewness in the trade-level return distribution . This pattern precisely the one observed in KSE Sentinel's own results (§5.3) is consistent with the behavioural explanation that "price trends exist in part due to long-standing behavioural biases exhibited by investors, such as anchoring and herding," as well as non-profit-seeking participants such as central banks . Hurst, Ooi, and Pedersen observe that "most bear markets have historically occurred gradually over several months, rather than abruptly over a few days, which allows trend followers an opportunity to position themselves short after the initial market decline and profit from continued market declines" .

2.3 Backtest Overfitting and Selection Bias

Bailey, Borwein, López de Prado, and Zhu (2014) formalize the risk that searching over many strategy variants inflates the apparent significance of the best-performing one the core motivation for this paper's use of the Deflated Sharpe Ratio . As Bailey and López de Prado (2014) state, "With the advent in recent years of large financial data sets, machine learning, and high-performance computing, analysts can backtest millions (if not billions) of alternative investment strategies. Backtest optimizers search for combinations of parameters that maximize the simulated historical performance of a strategy, leading to backtest overfitting" . The Deflated Sharpe Ratio "corrects for two leading sources of performance inflation: Selection bias under multiple testing and

non-normally distributed returns" helping "separate legitimate empirical findings from statistical flukes".

2.4 Algorithmic Trading Regulation in Frontier Markets

SECP's 2025 Concept Paper situates Pakistan within a broader global trend of regulators requiring registration, audit trails, senior management oversight, and killswitch capability for algorithmic trading systems . The proposed framework "draws on international best practices and clearly defines the responsibilities of key participants" , with exchanges responsible for registration and surveillance, brokerages required to implement internal controls, and third-party providers expected to meet compliance standards . The paper recommends a phased implementation, with initial access limited to institutional investors a direction KSE Sentinel's operational controls (§8) are built proactively toward.

Suggested full citations:

- Jegadeesh, N., and Titman, S. (1993). "Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency." *The Journal of Finance*, 48(1), 65–91.
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3. Market and Data

3.1 The Pakistan Stock Exchange

PSX's KSE-100 index composition and liquidity characteristics define the trading environment. The April 2024 one-share lot size reform replacing the prior 100-share lot standard is directly relevant to position sizing methodology (§4.3). The removal of UNITY from the KSE-100 as of April 1, 2026, exemplifies routine index composition changes the strategy's universe management must handle.

3.2 Universe

The strategy's tradeable universe comprises 30 symbols spanning PSX's major sectors (commercial banks, oil & gas exploration/marketing, fertilizer, power generation, cement), defined and version-controlled in the codebase (bulk_sync_universe.py).

Data note: Engro Corporation Limited (ENGRO) was formally delisted from PSX effective January 14, 2025, following a Scheme of Arrangement merging it into Dawood Hercules Corporation Limited, subsequently renamed Engro Holdings Limited (ENGROH). ENGROH remains actively traded but is not currently available through the strategy's primary data vendor under any ticker suffix attempted. This is disclosed transparently as a known, understood, immaterial-scale data gap.

3.3 Data Sources

Primary window (2021–2026): yfinance daily OHLCV data.

Independent out-of-sample validation window (~2000–2020): a separate PSX historical archive, used specifically to check whether the strategy's edge holds in a

dataset entirely disjoint from the one used for the primary backtest and any parameter selection.

3.4 Transaction Costs

All reported returns are net of an assumed 22.5 basis points in round-trip trading costs per leg (commission, slippage, and spread combined).

4. Strategy Methodology

Per standard convention, exact calibrated parameters, precise entry/exit trigger logic, and full source code are withheld here. This section describes the *mechanism class* and *validation discipline* in sufficient detail for methodological assessment, without functioning as a replication recipe.

4.1 Core Signal Logic

- A trend-continuation signal derived from short-horizon candle-close direction relative to a short-period exponential moving average, conditioned on a separately-identified prevailing trend regime.
- Exit logic strictness is conditional on whether the master trend regime has flipped since entry winners and losers are managed differently depending on whether the broader trend context that justified the entry remains intact.

4.2 Why Mechanization, Not Machine Learning

This is a **rule-based system**, not a machine-learned or "AI" model a deliberate design choice. Deterministic, auditable rules are both easier to validate statistically (no risk of an opaque model overfitting to noise in ways that resist detection) and more defensible to a regulator or institutional risk committee that needs to understand *why* a system does what it does, a concern SECP's Concept Paper explicitly raises.

4.3 Position Sizing

A tiered risk-based sizing framework (conservative/moderate/balanced/growth tiers, roughly 1–5% risk per trade) applied against a fractional-share-aware, per-symbol lot size resolver reflecting the actual post-2024 PSX one-share lot reform.

4.4 Portfolio-Level Risk Controls

- Correlation-based position gating: new entries are blocked when the candidate position would be highly correlated ($\rho \geq 0.70$) with an already-open position, computed from live synced price data.
- Sector/symbol concentration limits at the tenant configuration level.

5. Empirical Results

Results are from the canonical backtest *backtest_20260707_063609* cited by run ID for reproducibility and auditability.

5.1 Headline Statistics

Metric	Value
Total trades	1,009
Universe	~30 symbols
Sample window	~5 years (2021–2026)
Win rate	39.15%
Profit factor	2.51
CAGR (PKR)	7.47%
CAGR (USD-converted)	−2.96%
Sharpe ratio	2.08
Maximum drawdown	−2.91%
Monte Carlo significance (sign-permutation)	p = 0.0

Currency divergence: A positive local-currency return alongside a negative USDconverted return over the same period is a real, first-order finding for any foreign institutional allocator evaluating this strategy. It says as much about PKR currency risk over the sample window as it does about the strategy itself, and separating the two clearly is the kind of analytical honesty that distinguishes a credible practitioner paper from a sales document.

5.2 Win Rate vs. Expectancy

A sub-40% win rate alongside a strongly positive profit factor is not a contradiction it is the textbook signature of a well-functioning trend-following system: cut losses early and consistently, let winners run considerably longer. Average-win-to-averageloss ratio is approximately 3.9x, connecting directly to the Jegadeesh/Titman and Moskowitz-Ooi-Pedersen literature's characterization of momentum/trend strategy payoff structure.

5.3 Trade Return Distribution: Skewness and Kurtosis

Positive skew a small number of very large winning trades contributing disproportionately to total profit is consistent with established trend-following literature . This is not an idiosyncrasy specific to this strategy, but a signature shared with trendfollowing strategies documented across a century of data.

5.4 Benchmark Comparison

Performance against an equal-weight buy-and-hold benchmark over the same window and universe is reported, including comparisons that may be unflattering on some dimensions (e.g., raw USD return). Selective benchmarking is avoided.

.5 Sizing Sensitivity: 3% vs. 5% Risk Per Trade

We report a direct comparison against a 5% risk-per-trade variant, run on the identical universe, signal logic, and cost assumptions as the 3% canonical result only the position sizing tier differs.

Metric	3% Risk (canonical)	5% Risk
Total trades	1,009	602
Win rate	39.15%	39.37%
Sharpe ratio	2.08	2.09
CAGR (PKR)	7.47%	5.53%
Maximum drawdown	−2.91%	−2.68%

Observation: The 5% variant produced substantially fewer trades (602 vs. 1,009) than the 3% variant, despite identical entry/exit signal logic and, counterintuitively, a *lower* CAGR despite larger position sizing. Win rate and Sharpe ratio are essentially unchanged. The most plausible explanation is that larger position sizes interact with portfolio-level risk controls (correlation-based position gating, capital allocation limits) in ways that block entries a

smallersized book would have taken the sizing tier is not a free multiplier on the same trade sequence, but changes which trades the portfolio-level risk engine actually allows.

At the instrument level, the single best-performing symbol in the 5% run was NPL, returning 74.64% over the sample window on an isolated, independently-capitalized basis a strong individual result, but explicitly *not* representative of portfolio-level performance, which is properly reported as the 5.53% CAGR figure above.

6. Statistical Validation and Robustness

6.1 Monte Carlo Significance Testing

Sign-permutation methodology yields p = 0.0. This tests whether the observed sequence of wins/losses could plausibly have arisen from a random reordering of the same trade outcomes not whether the strategy's rules themselves were free from datasnooping in their original construction.

6.2 Walk-Forward Validation

Walk-forward validation of the adaptive sizing methodology shows out-of-sample Sharpe held in both folds tested. Two folds is suggestive, not definitive proof of robustness, and is described as such.

6.3 Deflated Sharpe Ratio - Addressing Multiple-Testing Bias Directly

The Deflated Sharpe Ratio methodology, per Bailey and López de Prado (2014), corrects a standard Sharpe ratio estimate for (a) the number of independent strategy trials/variants considered before arriving at the reported one, and (b) non-normality (skewness, kurtosis) in the return distribution . Both factors mechanically inflate an *uncorrected* Sharpe ratio's apparent significance.

The specific trial count used is N=4 (the 1%/3%/4%/5% sizing candidates). **This is a documented floor, not an exhaustive count.** Real prior experiments (e.g., a 5.5% sizing variant, an earlier stricter entry rule, an RSI-oversold gate, two "trap-immune" signal variants) are excluded specifically because they were abandoned before producing a comparable Sharpe figure to include not because they were forgotten. This is stated plainly, without implying N=4 is the complete history of strategy iteration.

The resulting Deflated Sharpe Ratio figure is [to be confirmed from `run_fresh_dsr_analysis.py output`].

6.4 Independent Historical Archive Validation

Testing against the separate ~2000–2020 PSX historical archive (§3.3) provides a genuinely distinct data source from the one used to develop and primarily test the strategy a stronger form of out-of-sample evidence than a simple time-split on a

single data source.

7. Live Forward Validation

This section is a genuine differentiator few practitioner papers can point to an actual, currently-running live forward test rather than only backward-looking simulation.

7.1 Program Design

- 60-day live forward-paper validation program, initiated July 6, 2026, running once per trading day.
- Every scan decision (executed, blocked, ignored, held, closed) persists to a dedicated live signal tracking table with exact entry-to-outcome linkage.

7.2 Real Operational Incidents Reported Transparently

These are evidence of engineering rigor and honest self-monitoring, not embarrassments to omit:

- A tenant risk configuration row was missing for the first two days of the live window, causing trades to be sized at a conservative fallback rather than the intended tier caught, diagnosed, and corrected, with the discontinuity documented.
- A timestamp-precision ID collision was caught live on Day 3 (two samemicrosecond events), safely contained by existing error handling, and fixed at the root cause the same day.

A research desk conducting due diligence will trust a paper *more*, not less, for disclosing that real bugs were found and fixed quickly during live operation this is precisely the operational maturity institutional allocators look for.

7.3 Interpreting Early Forward Results

A 60-day window, even fully clean, is not by itself sufficient to draw strong statistical conclusions. Its value at this stage is primarily as a genuine live-fire operational test of the infrastructure (signal generation, position sizing, risk gating, data integrity), not as a standalone performance claim.

8. Operational Infrastructure and Risk Management

- Multi-tenant architecture with per-tenant, per-timeframe risk configuration.
- Capacity analysis: an estimated ~PKR 500M hard capacity ceiling, with a practical operating range closer to ~PKR 100M at the recommended 4% sizing tier.
- Security and operational controls: rate limiting, CORS hardening, collisionsafe identifier generation, authentication/RBAC infrastructure, and a planned kill-switch mechanism explicitly aligned with SECP's stated regulatory direction.

9. Limitations

1. **Currency risk.** PKR-denominated returns are positive over the sample window; USD-converted returns are negative. Any foreign institutional capital allocating to this strategy bears PKR/USD exchange rate risk as a first-order consideration.
2. **Data gaps.** The ENGRO/ENGROH transition (§3.2) and any other known vendor coverage gaps are listed plainly.
3. **Sample size and multiple-testing.** The Deflated Sharpe Ratio's N=4 trial count is an honest floor, not an exhaustive history (§6.3).

4. **Frontier market liquidity and capacity.** The estimated capacity ceiling (§8) is a genuine constraint affecting which institutional allocators this strategy is actually suitable for.
5. **Regulatory environment still evolving.** SECP's algorithmic trading framework remains, as of this writing, a Concept Paper that has not been finalized into binding regulation .
6. **Not machine-learned; not a black box; also not infallible.** The strategy is a deterministic rule set derived from discretionary trading judgment. This provides interpretability advantages but is not a claim of superiority over other methodologies.

10. Conclusion

We have presented a mechanically-defined trend-continuation strategy for PSX, validated through backtesting, walk-forward analysis, an independent historical data source, and distinctively an actual live forward test with transparently-reported operational incidents. The strategy's 39.15% win rate, profit factor of 2.51, and Sharpe ratio of 2.08 are consistent with established trend-following literature , with positive skew and a small number of large winning trades driving the majority of profit.

Forward research directions: planned extension of the live validation window, planned kill-switch implementation aligned with emerging PSX/SECP regulatory direction, and research interest in porting the underlying methodology to other markets subject to independent data validation before any claims are made.

References

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